

STUDENT TESTIMONIAL & PROJECT SUBMISSION DOCUMENT

1. Student Information

Student Name	Adapa Uma Siva Sankar
Student ID	STU67ff0b8cbec7c1744767884
College	V.S.M. College Of Engineering, Ramachandrapuram
Branch & Year	B.Tech — CSE (Data Science) / 2026
Project Domain	Education & AI Developer Software Tools
Project Title	AI-Powered LaTeX TikZ Diagram Generator & Chatbot Studio
Submission Deadline	11 September 2026
GitHub Repository	https://github.com/ADAPAUMA/AI-LaTeX-TikZ-Studio

2. Executive Summary

This document serves as the official student testimonial for the **IBM AICTE Internship Program (Batch 2026)**. The selected problem statement addresses the difficulty academic researchers, professors, and engineering students face when manually authoring complex LaTeX TikZ diagram code for publication-grade figures. The solution, built **entirely with IBM Bob, IBM watsonx.ai, and IBM Watson Assistant**, provides an intelligent, conversational AI-driven diagram generation studio that removes this friction.

3. IBM Bob Project Implementation

The project was built exclusively using the **IBM Bob multi-agent orchestration platform**, leveraging the following IBM technologies:

IBM Bob Platform: Designed and orchestrated the multi-agent AI workflow — Chatbot Studio, TikZ Generator, Syntax Validator, Smart Fallback Engine, and Rendering Studio — using IBM Bob's visual pipeline builder.

IBM Watson Assistant (WebChat): Integrated a conversational AI agent (Integration ID: bad150a7-6816-40db-9249-10be96c432f4, Region: au-syd) into the Streamlit dashboard for natural-language query resolution and diagram guidance.

IBM Granite LLM: Used ibm/granite-13b-instruct-v2 via IBM watsonx.ai REST API for multi-agent reasoning, TikZ code synthesis, auto-correction, and diagram explanation generation.

Streamlit Web Dashboard: Built an interactive 5-tab AI studio with Watson WebChat integration, Granite AI Chatbot, pdflatex/pdf2svg vector rendering, and a Template Gallery.

GitHub Repository: All source code, documentation, requirements, and submission PPT are publicly accessible at: <https://github.com/ADAPAUMA/AI-LaTeX-TikZ-Studio>

4. Personal Testimonial

Working on this IBM AICTE Internship project has been one of the most rewarding and intellectually stimulating experiences of my academic journey. Building a full-stack AI studio from scratch — backed by IBM's enterprise-grade cloud infrastructure — has given me immense confidence and practical understanding of real-world AI application development.

The experience of working with IBM Bob's visual multi-agent orchestration, IBM Watson Assistant's NLU-powered WebChat, and IBM Granite's large language model capabilities has been truly transformative. I am sincerely grateful to IBM and AICTE for providing this incredible opportunity, and I look forward to continuing to build AI-powered solutions that make a meaningful difference in the academic community.

Student Signature:

Adapa Uma Siva Sankar
STU67ff0b8cbec7c1744767884

Institution:

V.S.M. College Of Engineering
Ramachandrapuram

Date:

10 September 2026
Deadline: 11 Sep 2026

This document was prepared in accordance with IBM AICTE Internship Program guidelines. | IBM Watson Assistant | IBM watsonx.ai | IBM Granite LLM | IBM Bob